



ADAMJEE GOVT. SCIENCE COLLEGE

PRELIMINARY EXAMINATION - 2026

Time: 3 Hours

CHEMISTRY- II

Max. Marks: 85

SECTION-A

Max. Marks: 17

(MULTIPLE CHOICE QUESTIONS)

Time: 20 minutes

NOTE: i) This section consists of 17 part questions and all are to be answered. Each part question carries one mark.

ii) Do not copy the part questions in your answerbook. Write only the answer in full against the proper number of the question and its part.

iii) The use of scientific calculator is allowed.

Choose the correct answer for each from the given options:

i. The alkali metal gives a yellow color during a flame test is:

*Na

*K

*Rb

*Cs

ii. The best oxidizing agent among halogens is:

*F₂

*Cl₂

*Br₂

*I₂

iii. A highly paramagnetic ion among the following is:

*Fe⁺²

*Fe⁺³

*Co⁺²

*Cr⁺³

iv. The element is not used for electroplating:

*Zinc

*Tin

*Chromium

*Manganese

v. Urea was first synthesized by Wohler from an inorganic material named as:

*Ammonium nitrate

*Ammonium chloride

*Ammonium cyanate

*Ammonium bicarbonate

vi. Formula of a saturated hydrocarbon is C₄H₈, it should be:

*Butane

*Butene

*Butyne

*Cyclobutane

vii. Which of the followings pairs of compounds represent functional group isomerism?

*1 - butene and 2 - butene

*Ethanol and dimethyl ether

*n - butane and isobutane

*Diethyl ketone and methyl propyl ketone

viii. Welding gas among the following is:

*Ethylene

*Acetylene

*Ethane

*Methane

ix. Amines act as bases because:

*they accept OH⁻

*they accept H⁺

*they donate H⁺

*they donate the OH⁻

x. Which of the following molecule cannot form hydrogen bonding with water molecule?

*Phenol

*Resorcinol

*Ethyl chloride

*Ethyl alcohol

xi. Which of the following is an anaesthetic agent?

*Phenol

*Ethyl alcohol

*Diethyl ether

*Acetone

xii. The reagent used to oxidized ketones into carboxylic acids is:

*Ammoniacal silver nitrate

*Potassium dichromate

*Fehling solution

*Benedict solution

xiii. Which of the following is not a synthetic plastic?

*Nylon

*Teflon

*Cellulose

*Polyethene

xiv. Drugs that lower the body temperature to normal are known as:

*Antibiotics

*Antipyretic

*Antiallergic

*Anti histamins

xv. The region of sphere which extend from 11 km to 50 km from our Earth's is known as:

*Troposphere

*Stratosphere

*Mesosphere

*Thermosphere

xvi. The primary goal of green chemistry is:

*Maximizing industrial production

*Minimizing waste generation

*Reducing energy consumption

*Developing environmentally friendly chemicals

xvii. The energy of IR radiations is:

*Higher than radio waves

*Higher than U.V waves

*Lower than micro waves

*Higher than x-rays

SECTION 'B'

(SHORT-ANSWER QUESTIONS)

Marks : 36

Note: Answer any Nine part questions. Select Four part questions from Inorganic – General Chemistry and Five part questions from Organic Chemistry. All part questions carry equal marks.

INORGANIC CHEMISTRY

i. What is meant by a diagonal relationship? Mention three pairs of representative elements that show diagonal relationship.

OR Define d-block elements, why do they form colored compounds? Explain it in term of Crystal Field Theory.

ii. Discuss the group trend of atomic radii & electrical conductivity of alkali metal.

OR Give reasons for the followings:

* Why Cu⁺² ion is blue but Zn⁺² is colourless?

* Why chromium exist in 4s¹ 3d⁵ configuration but not 4s² 3d⁴?

* Boiling point of halogens increase down the group in the periodic table?

* Electronegativities of alkali metals decrease from Li to Cs?

iii. Give the IUPAC names of the following complexes: (any four)

* Na₂[Pt(OH)₄]

* K₂[Fe(CN)₅NO]

* [Zn(NH₃)₄]⁺²

* [Ni(SCN)₄]⁻²

* [Pt(NH₃)₂Cl₂Br₂]

iv. Write down three properties of beryllium that show its unique behavior in group IIA.

v. What information about the structure of a molecule we can get from mass spectroscopy? Give the applications of mass spectroscopy.

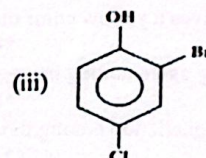
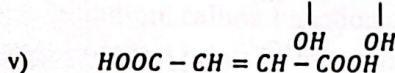
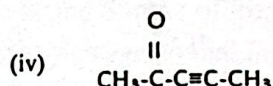
vi. Define green house gases. How green house gases cause to global warming?

vii. Write the names of two synthetic and two natural polymers.

OR What is antihistamine drug? Give the symptoms in which it is used.

ORGANIC CHEMISTRY

- viii. Define the following terms: (Any four)
 (a) Peptide bond (b) Catenation (c) Polymerization
 (d) Reforming of petroleum (e) Functional group
- OR**
 Define Bucky ball. Explain its structure and write two properties.
- ix. Two organic compounds A and B are given below:
 (A) $CH_2 = CH_2$ (B) $CH \equiv CH$
 * Draw the orbital structure of compound A and specify:
 * Hybridization * Bond angles
 * Write the ozonolysis reactions of B under suitable conditions and Baeyer test of A.
- OR**
 Give the equations and write the name of the final product in the following process.
 * Ethandiol is treated with HIO_4 .
 * 1,2-dibromoethane is heated with alcoholic KOH.
 * Ethene is treated with oxygen in presence of peracetic acid at $100^\circ C$.
 * Ethyne is treated with H_2O in the presence of $H_2SO_4/HgSO_4$.
- x. Write the IUPAC name of the following organic compounds: (Any four)



- xi. What are electrophilic substitution reactions? Write the mechanism of Friedel-Crafts alkylation or halogenations reaction of benzene.
- OR**
 Why are amines basic in nature? How can we prepare ethyl amine from the following compounds?
 * ethyl iodide * methyl cyanide * ethanamide
- xii. Identify each of the following with one laboratory test:
 * Alcohol * Phenol * Alkyl halide * Aldehyde
- OR**
 Define chiral carbon and plane polarized light. Explain optical isomerism briefly.
- xiii. What are organometallic compounds? How is Grignard reagent prepared? Write down the reactions of Grignard reagent with:
 $CH_3 - CH_2 - NH_2$ * CO_2
- OR**
 How can you prepare the following compounds from ethyl chloride?
 * Ethyl nitrile * Ethyl acetate * ethyl mercaptan * diethyl ether
- xiv. Name four derivative of carboxylic acid and write the equations of the following conversion.
 * carboxylic acid into acid anhydride * alkyl nitrite into carboxylic acid
 * acetic acid into acetyl chloride

SECTION 'C'

(DETAILED-ANSWER QUESTIONS)

Marks : 32

NOTE: Answer any Four questions — Two question from Inorganic – General Chemistry and Two from Organic Chemistry. All questions carry equal marks.

INORGANIC – GENERAL CHEMISTRY

3. Draw a flow diagram of contact process and describe various steps involved in the manufacturing of sulphuric acid.
4. Write the balance chemical equation for the following
 * Sodium metal put into ethyl alcohol * Copper is treated with conc. Nitric acid
 * Action of alkali on aluminium metal * Silicon react with steam
 * Bromine passes into water
 * Reaction between $KMnO_4$ and $FeSO_4$ in the presence of H_2SO_4 (write ionic equation)
 * Reaction of manganese with dilute sulphuric acid
 * Reaction of permanganate with oxalic acid
- OR**
 Describe how 99.99% pure copper is obtained from its chalcopyrite ore.
5. Define Troposphere and stratosphere. Describe the chemistry involved due to the presence of oxides of carbon and nitrogen in the troposphere. (Write equations where necessary)

ORGANIC CHEMISTRY

6. What is meant by orientation in Benzene? What are ortho, para and meta director group? Give example for each. Give the equation for the preparation of m-nitrotoluene and TNT from benzene.
- OR**
 Explain Molecular orbital treatment (M.O.T.) of Benzene.
7. Draw the structure of the following organic molecules:
 * isopropylacetylene * p-cresol * terephthalic acid
 * Glyoxal * β -methyl valer aldehyde * Ethyl neo-pentyl ether
 * Hydroquinone * iso-butylformate
- OR**
 Write the equation for the following conversion:
 * propanol into propanone * acet aldehyde into ethane
 * carbolic acid into picric acid * ethylene glycol into formaldehyde
 * acetic acid into ethanol * toluene into benzoic acid
 * ethyl chloride into vinegar acid * formaldehyde into imine
8. What is β -elimination? Discuss the mechanism of E_1 and E_2 reactions.
- OR**
 What are Carbohydrates? Classify them on the basis of structure and give their biological significance.